

Loan Approval: Financial Risk Analysis

A study of 20,000 loan applications: which applicant details are linked to a loan being approved, and which ones make little difference

Applications	Approved	Approval rate	Strongest factor	Model accuracy
20,000	4,780	23.9%	Annual income	88.3%

Tools: Python, pandas, seaborn, matplotlib and scikit-learn

Notebook, data and README:

github.com/nachiket-1/Loan-Approval-Financial-Risk-Analysis

This PDF is a readable copy of the whole project for phones and tablets. It follows the notebook from start to finish: the data, the checks, the findings for every variable, the prediction model, the model checks (cross validation, significance tests and a fairness check), what a lender should take from the results, and the conclusions. The code is in the notebook and the main parts are listed in Appendix B.

1. Overview

The goal of this project is to find out which applicant characteristics are linked to a loan being approved, and which are not. Every variable is studied the same way: applicants are split into groups, and the share of each group that gets approved (the approval rate) is compared.

Only about one application in four is approved (23.9%). Income decides most of it. The highest earning quarter of applicants is approved about 65% of the time, the lowest earning quarter about 1% of the time. The size of the loan compared with income, and the share of income already going to debt, matter almost as much. Savings balance and job tenure make almost no difference.

A logistic regression built only from applicant information predicts 88.3% of decisions correctly on applications it has never seen (88.0% in 5 fold cross validation), against 76.1% for a model that rejects everyone. A random forest does no better, and the model was also checked for fairness across age groups and other groups (section 6).

2. Dataset and preparation

The data comes from the Financial Risk for Loan Approval dataset on Kaggle. The file `Loan.csv` has 20,000 applications and 36 columns, with no missing values and no duplicate rows.

The dataset is synthetic. The application dates run one per day from January 2018 to October 2072, so there is no real trend over time. The dates are only used in section 4.1 to confirm this, and not in the rest of the analysis.

Checks done on the data

1. Data types were checked. `ApplicationDate` was stored as text and was converted to a date type.
2. Missing values were counted: none.
3. Duplicate rows were counted: none.
4. Eleven columns of interest were kept in a separate copy of the data, so the original stays untouched.

Columns used

Column	Meaning
AnnualIncome	Yearly income of the applicant
CreditScore	Credit score (343 to 712 in this data)
EmploymentStatus	Employed, Self-Employed or Unemployed
LoanAmount	Amount requested
SavingsAccountBalance	Money in the savings account
TotalAssets, TotalLiabilities	What the applicant owns and owes
JobTenure	Years in the current job
HomeOwnershipStatus	Mortgage, Own, Rent or Other
LoanApproved	The result: 1 approved, 0 rejected
ApplicationDate	Date of the application (not used for findings)
TotalDebtToIncomeRatio, InterestRate, RiskScore	Added later for a closer look
LoanToIncome (new)	Loan amount divided by annual income
AssetLiabilityRatio (new)	Total assets divided by total liabilities

3. How approval was measured

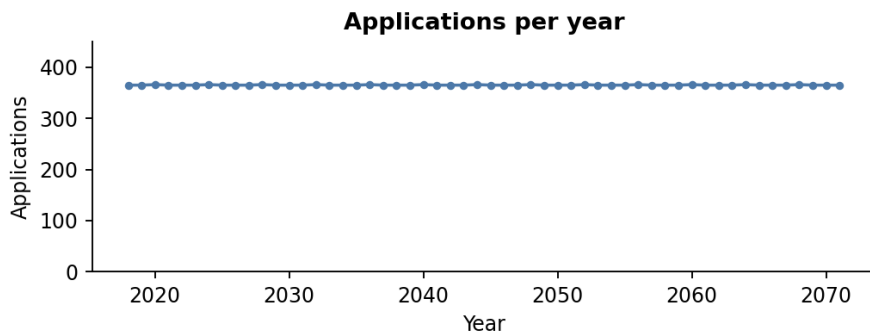
Counting how many approved loans each group has is misleading, because a bigger group always has more approvals. This project compares the **approval rate** instead: approved loans divided by applicants in the group.

Numeric variables are split into four groups of equal size (quartiles), so every group holds about 25% of the applicants. Credit score, the assets to liabilities ratio and job tenure use fixed bands that make sense for that variable. In every chart the red dashed line marks the overall approval rate of 23.9%. A group above the line is approved more often than average, a group below it less often.

4. Findings

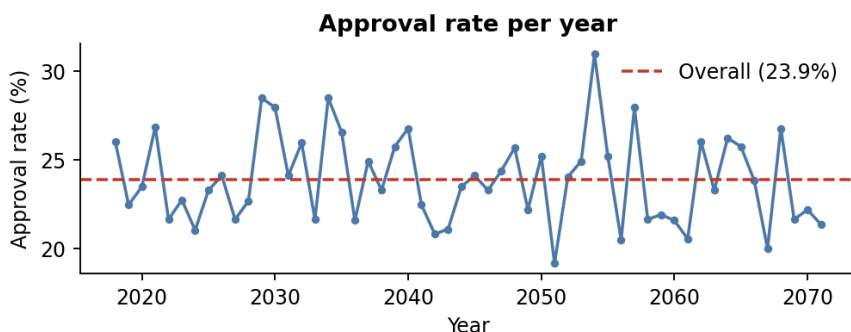
4.1 Applications over time

The data has exactly one application per day, so every full year has about 365 applications. The last year (2072) is incomplete and is left out of the charts.



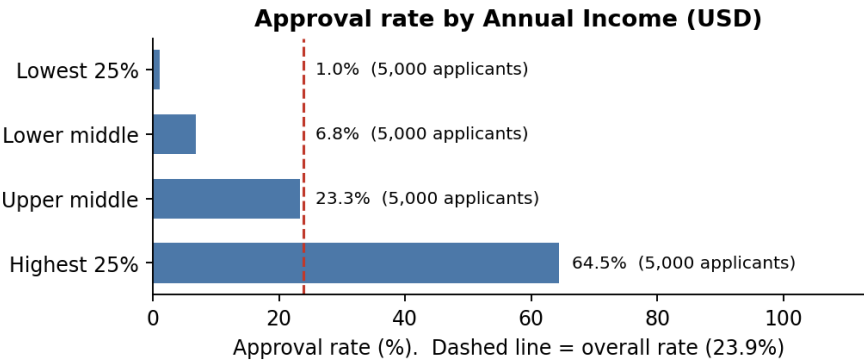
Applications per year. The line is flat because the dates are artificial.

The yearly approval rate moves between roughly 19% and 31% with no upward or downward trend, so time is not useful for explaining approval. It is dropped from the rest of the analysis.



Approval rate per year compared with the overall rate.

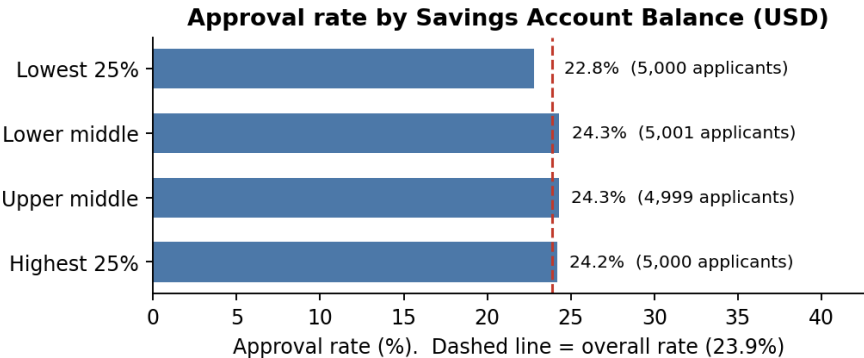
4.2 Annual income



Four equal income groups of 5,000 applicants each.

Income is the strongest applicant factor. The lowest quarter of earners (up to about USD 31,700 a year) is approved about 1% of the time, the next about 7%, the next about 23%, and the highest quarter (above about USD 74,400) about 65% of the time. The highest earners make up 67% of all approved loans (3,223 of 4,780).

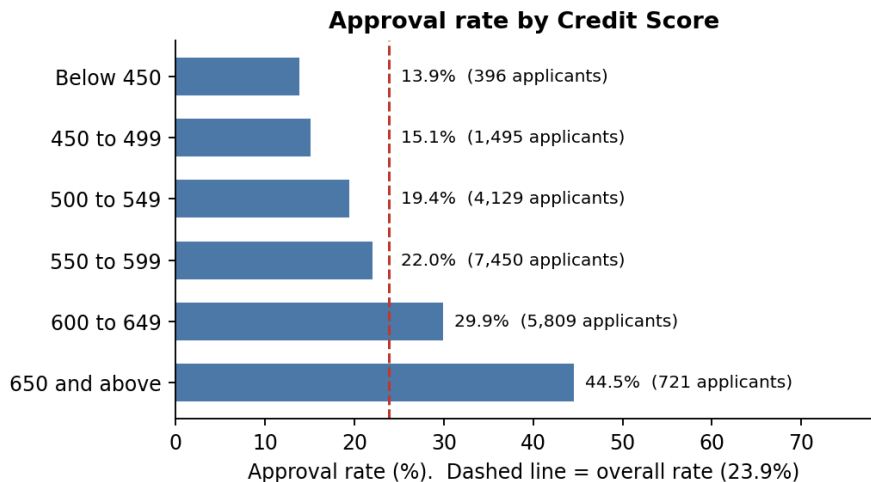
4.3 Savings account balance



Savings balance in four equal groups.

Savings balance has almost no effect. All four groups sit between 22.8% and 24.3%. The distribution is skewed: the average balance is about USD 4,950 but the median is about USD 2,990, because a few very large balances pull the average up.

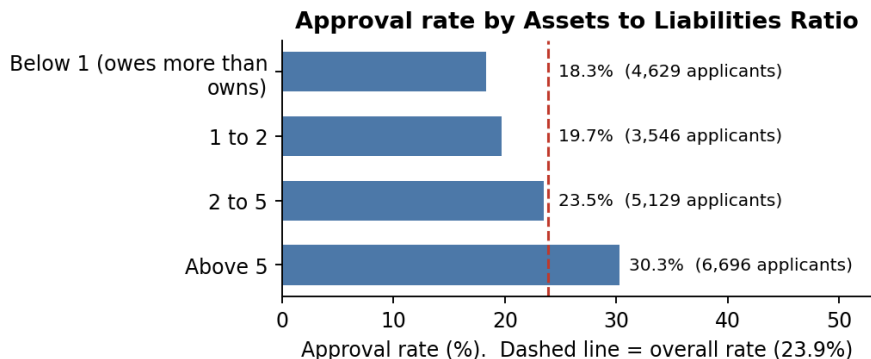
4.4 Credit score



Credit scores in 50 point bands.

Credit scores in this data run from 343 to 712, so the usual lender bands such as Very Good (740 and above) would be empty. Fifty point bands are used instead. The approval rate rises steadily, from about 14% below 450 to about 45% for 650 and above. Most applicants sit between 550 and 649 (13,259 of 20,000), so the best and worst bands are small. Credit score matters, but much less than income.

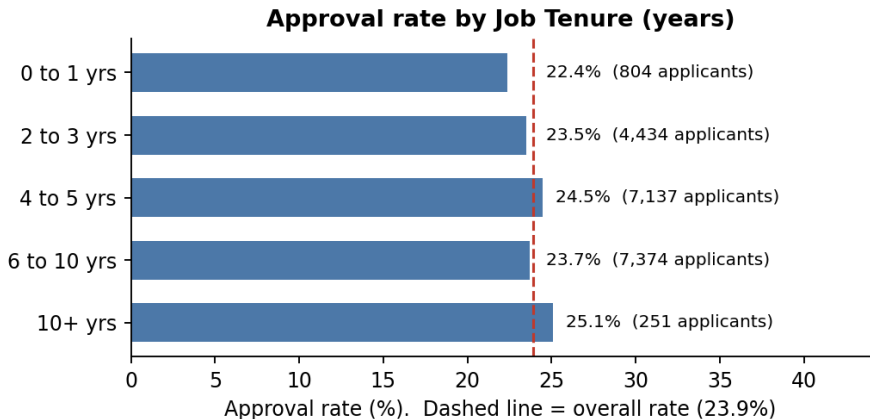
4.5 Assets to liabilities ratio



Total assets divided by total liabilities.

Applicants with more assets compared with what they owe are approved more often, from 18.3% when they owe more than they own to 30.3% when assets are more than five times liabilities. The ratio is very skewed (average about 7.2, median about 2.7, maximum above 550), so the median describes a typical applicant better. The effect is real but moderate.

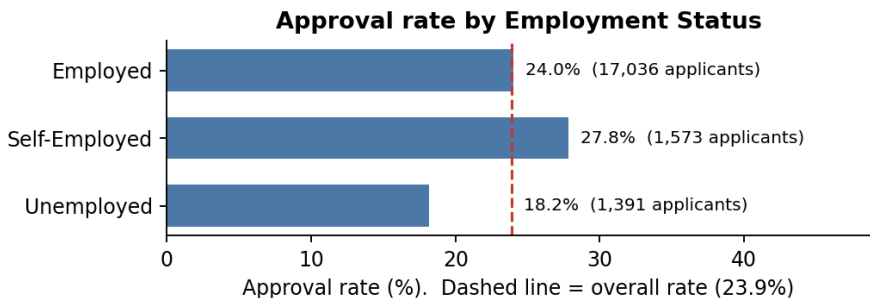
4.6 Job tenure



Years in the current job.

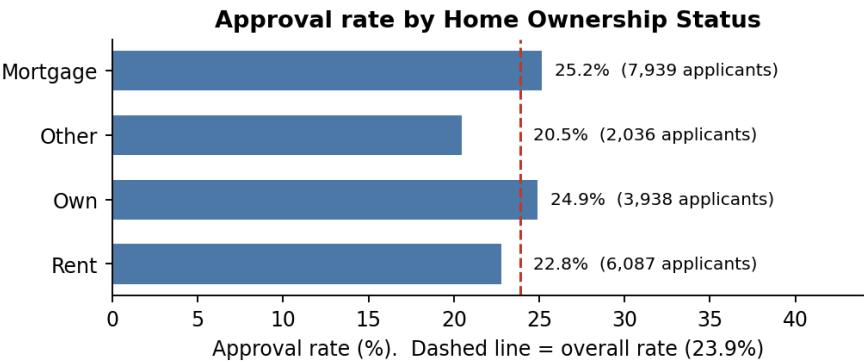
Job tenure makes almost no difference. Approval rates range only from 22.4% to 25.1%. The 0 to 1 year and 10+ year groups are small (804 and 251 applicants), so their rates are less reliable.

4.7 Employment status and home ownership



Approval rate by employment status.

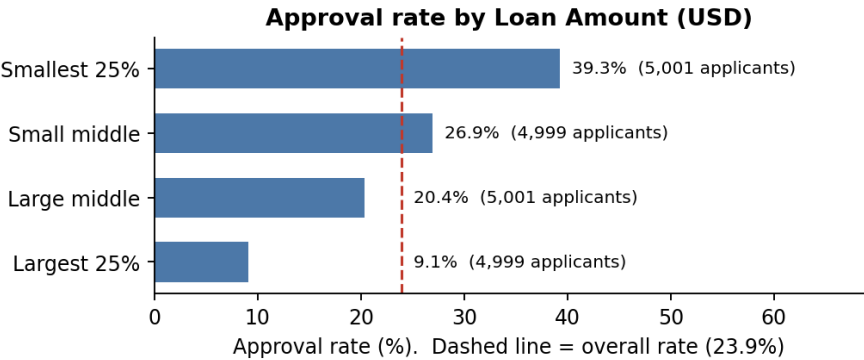
Unemployed applicants are approved least often (18.2%). Self-employed applicants are slightly above average (27.8%). Employed applicants are close to average (24.0%) and make up most of the data (17,036 applicants).



Approval rate by home ownership status.

Home ownership has a small effect. Applicants with a mortgage (25.2%) or who own their home (24.9%) are approved slightly more often than renters (22.8%) and the Other group (20.5%).

4.8 Loan amount

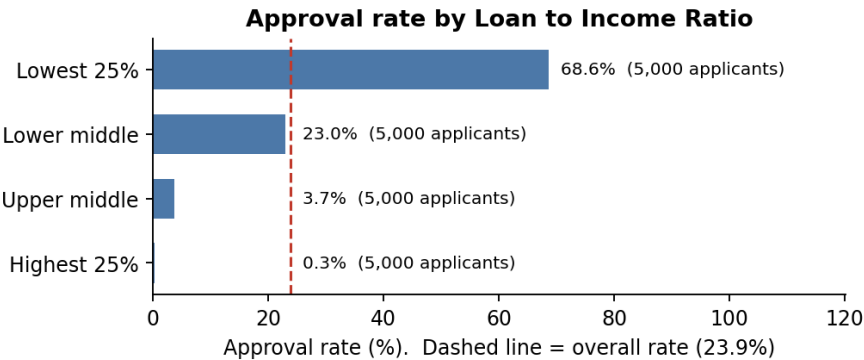


Loan amount in four equal groups.

The bigger the loan, the lower the approval rate: 39.3% for the smallest quarter of loans down to 9.1% for the largest quarter. Loan amount is the second strongest variable after income, and it works in the opposite direction.

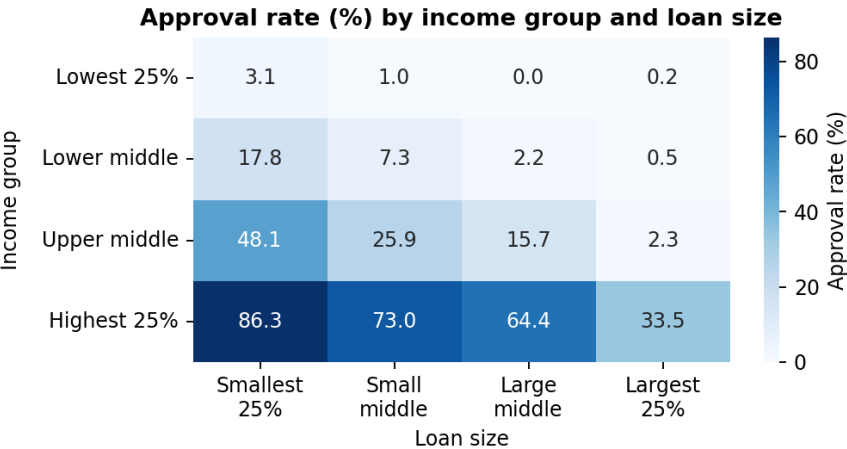
4.9 Loan amount compared with income

A loan of USD 30,000 is easy for a high earner and hard for a low earner, so a new column was created: loan amount divided by annual income. A value of 0.5 means the loan is half of the yearly income.



Loan to income ratio in four equal groups.

The ratio separates applicants very sharply. When the loan is up to about 26% of yearly income, about 69% of applications are approved. When it is above about 78% of yearly income, only 0.3% are (16 out of 5,000).

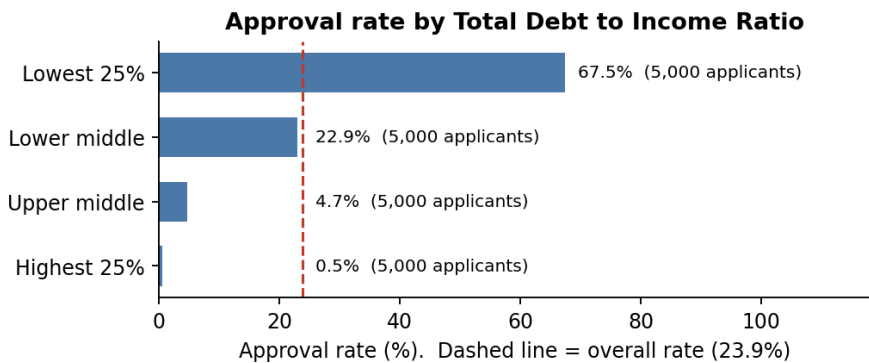


Approval rate for every combination of income group and loan size.

Low earners are almost never approved, even for the smallest loans (about 3%). High earners are approved 86% of the time for small loans but only 34% of the time for the largest ones. Loan size matters most for people who earn enough to be considered at all.

One correction to an early guess: the ratio does not beat income on its own in a simple correlation (about -0.41 against about 0.60), because it is very skewed and the link is not a straight line. It is still a much sharper divider than loan amount alone.

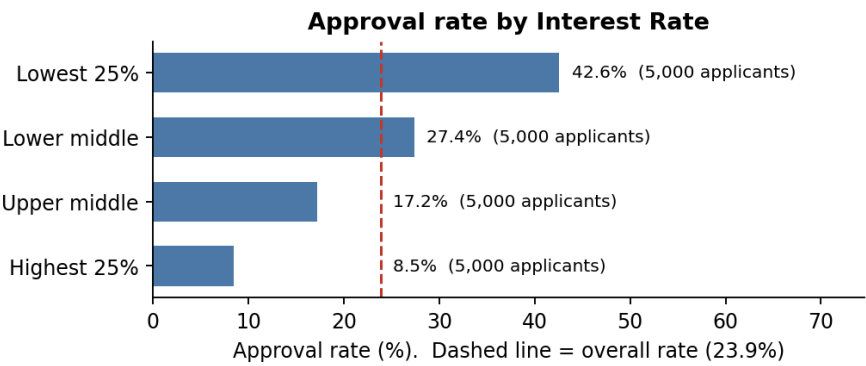
4.10 Debt to income ratio



Total debt to income ratio in four equal groups.

The less of their income an applicant owes, the more likely approval is. The lowest quarter (ratio up to about 0.18) is approved 67.5% of the time and the highest quarter (above about 0.51) only 0.5% of the time. This is a genuine applicant characteristic. It overlaps with income, which is expected because income is part of the ratio.

4.11 Interest rate

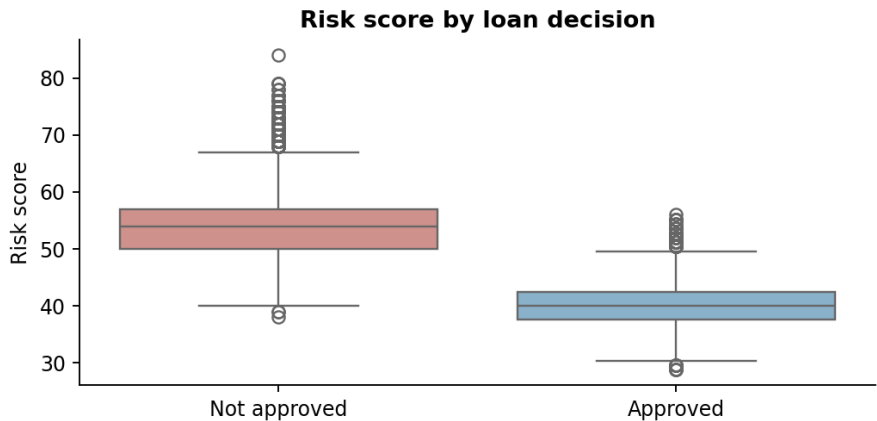


Interest rate in four equal groups.

Approval falls from 42.6% for the lowest interest rates to 8.5% for the highest. The interest rate is set by the lender and riskier applicants are usually offered higher rates, so it partly reflects the lender's view of the applicant. It was studied but kept out of the prediction model.

4.12 Risk score

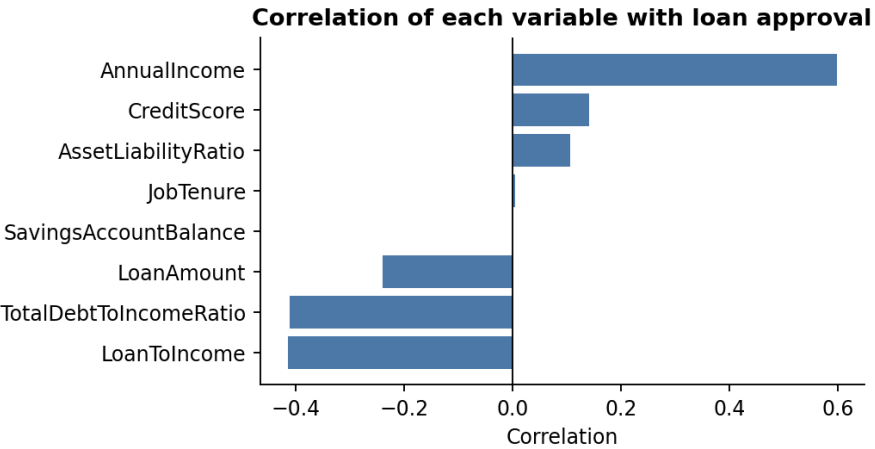
RiskScore has the strongest link to approval of any column (correlation about -0.77). Approved applicants have a clearly lower score (average about 40) than rejected ones (average about 54), and the two groups barely overlap.



Risk score for rejected and approved applications.

The score is most likely calculated from the same information the lender uses to make the decision. Using it to explain approval would be circular, like predicting exam results from the marks given by the examiner. It is therefore kept out of the main model.

4.13 All numeric variables together



Correlation with loan approval. Correlation shows association, not cause.

Variable	Lowest approval rate	Highest approval rate	Effect
Annual Income (USD)	Lowest 25% (1.0%)	Highest 25% (64.5%)	Very strong
Loan to Income Ratio	Highest 25% (0.3%)	Lowest 25% (68.6%)	Very strong
Total Debt to Income Ratio	Highest 25% (0.5%)	Lowest 25% (67.5%)	Very strong
Loan Amount (USD)	Largest 25% (9.1%)	Smallest 25% (39.3%)	Strong
Interest Rate	Highest 25% (8.5%)	Lowest 25% (42.6%)	Strong
Credit Score	Below 450 (13.9%)	650 and above (44.5%)	Moderate
Assets to Liabilities Ratio	Below 1 (owes more than owns) (18.3%)	Above 5 (30.3%)	Moderate
Employment Status	Unemployed (18.2%)	Self-Employed (27.8%)	Small
Home Ownership Status	Other (20.5%)	Mortgage (25.2%)	Small
Job Tenure (years)	0 to 1 yrs (22.4%)	10+ yrs (25.1%)	None
Savings Account Balance (USD)	Lowest 25% (22.8%)	Lower middle (24.3%)	None

5. Predicting approval

A logistic regression was trained to predict whether a loan is approved. It was chosen because it gives a probability of approval and its coefficients are easy to read.

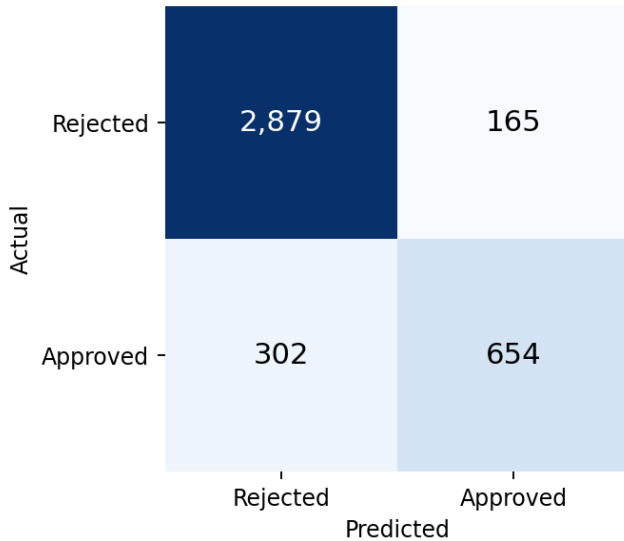
Features: annual income, credit score, loan amount, savings balance, job tenure, assets to liabilities ratio, loan to income ratio, debt to income ratio, employment status and home ownership. RiskScore and InterestRate were left out because the lender sets them. The data was split 80% for training (16,000 applications) and 20% for testing (4,000 applications), keeping the same approval share in both parts. Variables were scaled so the coefficients can be compared.

Metric	Without RiskScore (used)	With RiskScore (comparison only)
Accuracy	88.3%	98.3%
Precision	79.9%	97.1%
Recall	68.4%	95.7%
F1 score	73.7%	96.4%
ROC AUC	0.941	0.998

The model is right 88.3% of the time on applications it has never seen. A model that rejects every application would be right 76.1% of the time. Accuracy on the training data was 88.0%, almost the same, so the model is not overfitting. The ROC AUC of 0.94 means it ranks approved applicants above rejected ones very reliably.

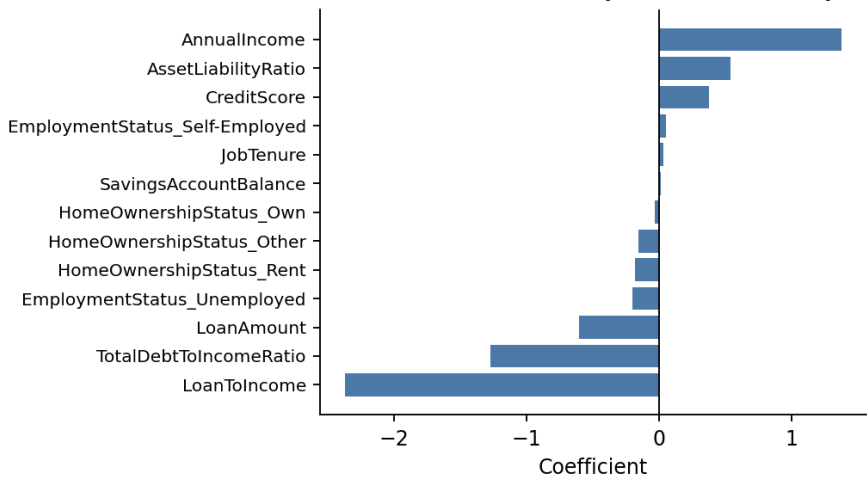
Of the loans the model predicts as approved, 79.9% really were (precision). Of the loans that really were approved, it finds 68.4% (recall). In the confusion matrix, 302 approved loans were missed and 165 rejected loans were wrongly predicted as approved.

Confusion matrix (4,000 test applications)



Confusion matrix on the test data.

Model coefficients (scaled variables)



Positive coefficients push towards approval, negative towards rejection.

The loan to income ratio is the strongest push towards rejection, followed by the debt to income ratio and the loan amount. Annual income, the assets to

liabilities ratio and credit score push towards approval. Savings balance and job tenure are close to zero, which matches the earlier findings. Income, loan amount and loan to income overlap, so the overall pattern matters more than any single coefficient.

Choosing the cutoff

The default cutoff is 0.5. Lowering it finds more approved loans but is less precise. The best choice depends on the cost of a bad loan compared with a lost customer.

Cutoff	Accuracy	Precision	Recall
0.3	87.1%	68.7%	84.1%
0.4	88.2%	74.6%	76.9%
0.5	88.3%	79.9%	68.4%

What happens with RiskScore

Adding RiskScore lifts accuracy from 88.3% to 98.3% and the ROC AUC from 0.94 to 0.998. That looks impressive but it is the circular effect described earlier: the score already contains the lender's judgement, so the model is partly copying the answer. The honest model is the one without RiskScore, and 88% accuracy from applicant information alone is a good result.

6. Checking the model further

A single train and test split can be lucky or unlucky. In this section the model is tested with 5 fold cross validation, compared with a random forest and compared with simpler versions of itself. After that the small differences found earlier are tested to see whether they are real, and the model is checked for fairness.

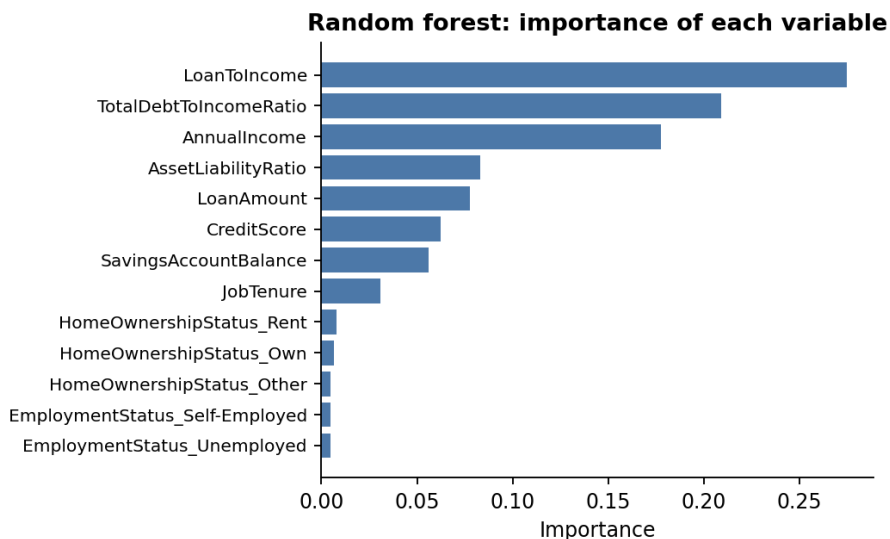
6.1 Cross validation and a second model

In 5 fold cross validation the data is split into 5 parts. The model is trained on 4 parts and tested on the fifth, five times over, so every application is used for testing exactly once. The result is an average with a spread (the number after +/-). A random forest is added for comparison because it can capture threshold effects that a straight line model cannot.

Metric	Logistic regression	Random forest
Accuracy	0.880 +/- 0.002	0.879 +/- 0.002
Precision	0.786 +/- 0.013	0.774 +/- 0.005
Recall	0.687 +/- 0.013	0.697 +/- 0.009
F1 score	0.733 +/- 0.005	0.733 +/- 0.005
ROC AUC	0.939 +/- 0.003	0.937 +/- 0.002

The random forest is no better than the logistic regression. Accuracy is 0.880 against 0.879 and ROC AUC is 0.939 against 0.937, and both gaps are smaller than the fold to fold spread. The simpler model is therefore the better choice, because it is easier to explain and its coefficients can be read.

The cross validated accuracy of 88.0% is slightly below the 88.3% from the single split in section 5, so 88.0% is the more honest figure to quote. The suggestion to try a tree based model was tested here and did not help on this data.



The random forest relies on the same variables as the logistic regression.

The random forest ranks the same variables at the top: loan to income ratio (about 0.28), debt to income ratio (about 0.21) and annual income (about 0.18). Employment status and home ownership come last. Savings balance and job

tenure get small but non zero scores, because a forest can always find some split in a continuous variable, so these scores should not be read as a real effect.

6.2 How simple can the model be?

A model that a lender can explain in one sentence is easier to trust and to check. The same cross validation was repeated with fewer variables.

Version of the model	Accuracy	Recall	ROC AUC
All 10 variables (current model)	0.880	0.687	0.939
Without employment and home ownership	0.880	0.688	0.938
Without savings and job tenure	0.881	0.689	0.939
Loan to income and debt to income only	0.863	0.692	0.921
Those two plus income and credit score	0.876	0.689	0.930

Removing employment status and home ownership costs nothing (accuracy 0.880 and ROC AUC 0.938, against 0.880 and 0.939). Removing savings balance and job tenure also costs nothing. A model that uses only the loan to income ratio and the debt to income ratio still reaches 86.3% accuracy, and adding income and credit score lifts it to 87.6%. Most of the predictive power sits in two affordability ratios.

6.3 Are the small differences real?

A difference between two groups can be chance. Two tools are used to check. The **chi-square test** gives a p-value, and a p-value below 0.05 means the difference is unlikely to be chance. With 20,000 applications even tiny differences pass this test, so it cannot be used alone. **Cramer's V** shows how big the difference is, from 0 (no link) to 1 (a perfect link). Roughly, below 0.1 is a very weak link, 0.1 to 0.3 moderate and above 0.3 strong.

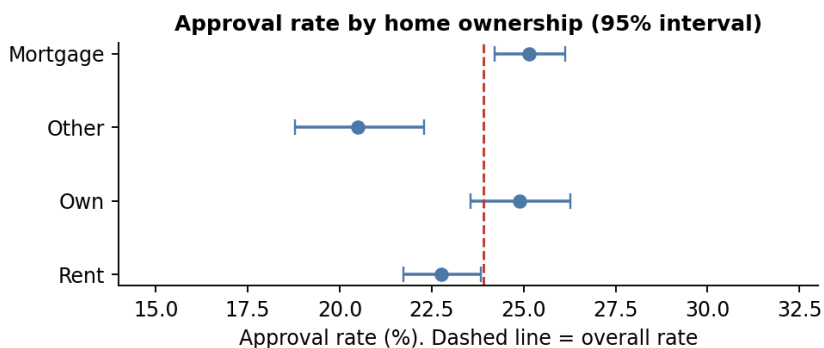
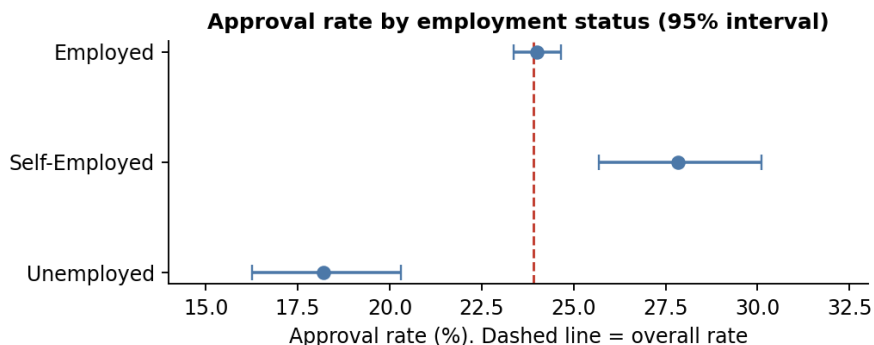
Variable	Chi-square	p-value	Cramer's V	Real?
Loan to income ratio	8,152.6	< 0.001	0.638	Yes
Debt to income ratio	7,742.9	< 0.001	0.622	Yes
Annual income	6,760.4	< 0.001	0.581	Yes
Loan amount	1,314.3	< 0.001	0.256	Yes
Credit score	428.6	< 0.001	0.146	Yes
Assets to liabilities ratio	262.4	< 0.001	0.115	Yes
Employment status	38.5	< 0.001	0.044	Yes
Home ownership	26.3	< 0.001	0.036	Yes
Savings balance	4.5	0.216	0.015	No
Job tenure	3.2	0.529	0.013	No

Eight of the ten differences are real and two are not. Savings balance (p about 0.22) and job tenure (p about 0.53) cannot be told apart from chance, which confirms what the charts suggested: they do not matter for approval.

Employment status and home ownership pass the test, but their Cramer's V is only 0.044 and 0.036, a very weak link. They are real but tiny. The strongest links are the loan to income ratio (0.64), the debt to income ratio (0.62) and annual income (0.58), then loan amount (0.26), credit score (0.15) and the assets to liabilities ratio (0.12).

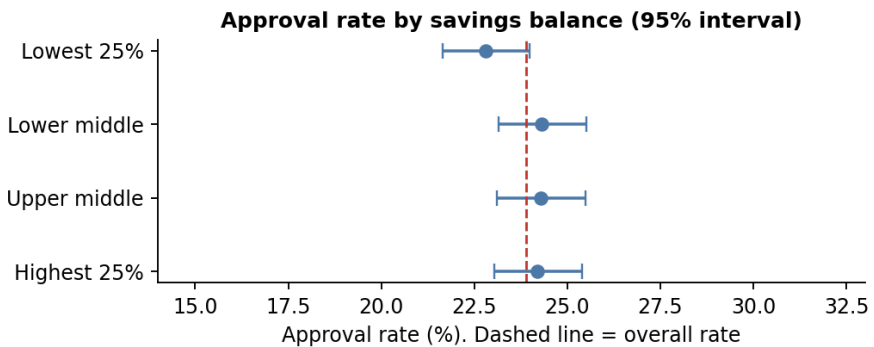
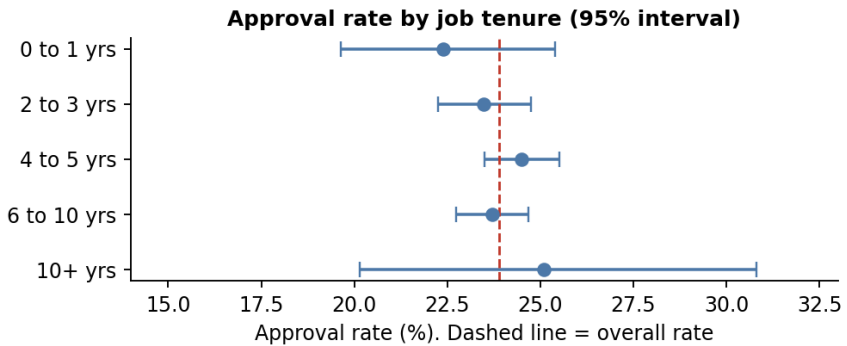
When the variables are split into groups, the loan to income ratio separates approved from rejected slightly better than income does. This fits the earlier note that it was a much sharper divider than loan amount, and adjusts the simple correlation result (about 0.41 against 0.60), which only measures straight line links.

A **95% confidence interval** shows the range in which the true approval rate of a group probably lies. If the ranges of two groups do not overlap, the groups really differ.



Approval rate with a 95% interval for employment status and home ownership.

Employment status shows real differences. The intervals for employed (24.0%, range 23.4 to 24.6), self-employed (27.8%, range 25.7 to 30.1) and unemployed applicants (18.2%, range 16.2 to 20.3) do not overlap. For home ownership, mortgage (25.2%) and own (24.9%) overlap each other, renters (22.8%, range 21.7 to 23.8) sit just below them, and the Other group (20.5%) is lowest.

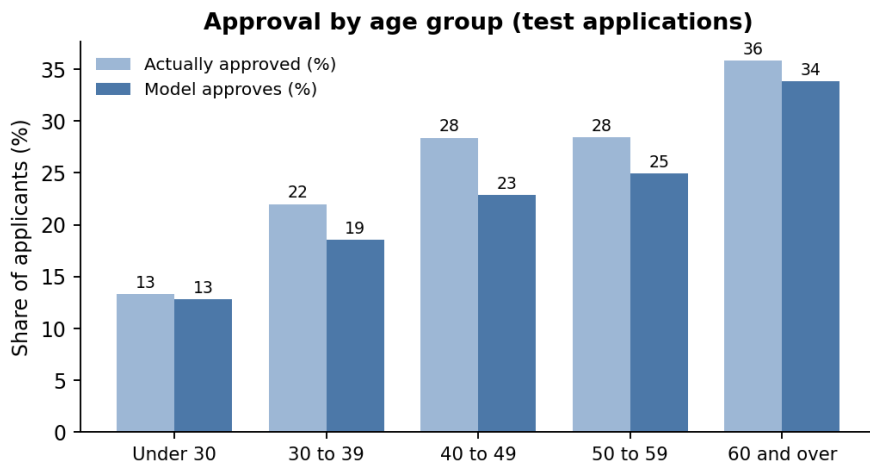


The same view for job tenure and savings balance.

For job tenure and savings balance every interval overlaps with every other. The 10+ year group has a very wide interval (20.1 to 30.8) because it has only 251 applicants. These two variables show no real difference between groups.

6.4 Fairness check

The model does not use age, marital status or education level. But other variables can act as stand ins for them. Younger applicants, for example, tend to have lower incomes and credit scores. This check asks whether approval, and the model's decisions, still differ across these groups. It looks at the 4,000 test applications only.



Share actually approved and share the model approves, by age group.

Age group	Applicants	Actually approved	Model approves	Approved loans found
Under 30	773	13.3%	12.8%	60.2%
30 to 39	1,227	22.0%	18.6%	64.1%
40 to 49	1,197	28.4%	22.9%	69.4%
50 to 59	605	28.4%	25.0%	73.8%
60 and over	198	35.9%	33.8%	78.9%

Approved loans found = the share of loans that were really approved which the model also approves.

Age is strongly linked to approval in the data, from 13.5% for applicants under 30 to 30.8% for those aged 60 and over. The model never sees age, yet its decisions follow the same pattern: it approves 12.8% of the under 30 test applicants against 33.8% of those aged 60 and over, and it finds 60% of the loans that were really approved for the younger group against 79% for the older group.

Age group	Median income	Median credit score	Median loan / income	Lowest income 25%	Highest income 25%
Under 30	41,550	558	0.52	0.5%	49.4%
30 to 39	46,599	572	0.47	1.0%	62.9%
40 to 49	50,972	585	0.43	1.5%	68.6%
50 to 59	55,088	598	0.41	1.2%	70.1%
60 and over	57,807	614	0.37	1.9%	66.2%

The last two columns show the approval rate inside the lowest and the highest income quarter.

Part of the gap is explained by the variables the model does use. Younger applicants have a lower median income (about 41,550 against 57,807 for those 60 and over), lower credit scores (558 against 614) and ask for larger loans compared with income (the median loan is about 52% of income for the under 30 group against 37% for those 60 and over). But it is not fully explained: within the highest income quarter, 49% of applicants under 30 are approved against 69% of those aged 40 to 49.

Marital status is not linked to approval in the data (p about 0.62, all groups between 22% and 25%), yet the model finds fewer approved loans among single applicants (62%) than among married ones (72%). This comes from test groups of about 1,250 and 2,000 applicants, so it is something to check further rather than a proven gap. The model also finds fewer of the approved loans among unemployed applicants (56% against 69% for employed) and renters (63% against 73% for those who own their home). Education level is strongly linked to approval in the data (14% for high school to 44% for doctorate) and the model approves less often than the data for the higher education groups. The tables for marital status, education, employment and home ownership are in Appendix C.

7. What a lender should take from these results

1. Affordability decides most outcomes. The loan to income ratio and the debt to income ratio on their own predict 86% of decisions correctly, and the full model reaches 88%. A lender can explain these two ratios to an applicant in plain words, which matters for trust and for any complaint or review.

2. Do not penalise what does not matter. Savings balance and job tenure make no real difference to approval in this data, and removing them from the model costs nothing. Rejecting someone for a short time in their job or a low savings balance is not supported by these results.

3. Use the model to support people, not to replace them. At the standard cutoff the model misses about 3 in 10 of the loans that were approved. Applications with a probability of approval between about 0.3 and 0.5 are good candidates for a manual review.

4. Treat fairness as an ongoing check. The model does not use age, marital status or education, but it still approves younger applicants and renters less often, partly because income and credit score differ between groups and partly for reasons these data cannot explain. A lender should repeat the group checks above on every new version of the model and look at both approval rates and the share of good applications found.

5. Remove variables that add almost nothing. Employment status and home ownership have a very weak link to approval (Cramer's V below 0.05) and the model is just as accurate without them. Dropping them lowers fairness risk at no cost in accuracy. Rules on which characteristics may be used in credit decisions differ from country to country, so this needs legal review before any real use.

6. Do not trust RiskScore or InterestRate to explain approval. Both are produced by the lender's own process. Using them would make the model look better (98% accuracy) while explaining nothing.

Limits of this project. The dataset is synthetic, so the sharp patterns, such as almost no approvals above a certain loan to income ratio, describe how the data was generated and not how real lenders behave. The model should be tested again on real application data before anyone relies on it. Statistical links in this project show association, not cause.

8. Improvements made along the way

The first version of the analysis was reviewed and corrected in these ways.

1. A typo changed the meaning of a note. The note under the approved applicants chart said the lowest earners had the "list approved" applications,

which reads as the most approved. The chart shows the opposite: the highest earners have the most approved loans and the lowest earners the fewest. The note was corrected.

2. **Income groups were uneven.** Using the mean as a cutoff made one group much wider than the others (7,522 applicants against 5,000 or fewer). Equal sized quartiles replaced it.

3. **Counts were replaced by approval rates**, since counts mostly show where the most applicants are.

4. **A pandas warning** (SettingWithCopyWarning) was removed by working on a proper copy of the data.

5. **The time chart** was corrected for the incomplete last year, and a note explains that the dates are artificial.

6. **Repeated code** was moved into three small helper functions, and the mode now prints as a single number.

7. **The model was checked further** after a review pointed out the missing significance tests and fairness check: cross validation, a random forest comparison, chi-square tests, confidence intervals and a group by group fairness check were added (section 6).

9. Conclusions and next steps

Conclusions

1. **Annual income is the strongest applicant factor.** The approval rate climbs from about 1% in the lowest income quarter to about 65% in the highest (correlation about 0.60).

2. **What matters is the size of the loan compared with income.** Approval is about 69% when the loan is up to a quarter of yearly income and almost 0% when it is above three quarters of it. Debt to income ratio tells the same story (67.5% against 0.5%). These two ratios alone predict 86% of decisions correctly.

3. **Loan amount works against the applicant**, falling from 39% for the smallest loans to 9% for the largest, but low earners are rejected even for small loans.

4. **Credit score, assets to liabilities ratio, employment status and home ownership have a moderate or small effect.** Higher credit scores, a higher ratio, self-employment and owning or mortgaging a home all go with higher

approval rates.

5. **Savings balance and job tenure have no real effect.** Chi-square tests confirm that the differences between their groups are not statistically significant (p about 0.22 and 0.53), and application date shows no trend.
6. **Interest rate and RiskScore are set by the lender**, so they were studied but not used to explain approval. A logistic regression with applicant information only predicts about 88% of test decisions correctly (88.0% in 5 fold cross validation, baseline 76.1%, ROC AUC 0.94).
7. **A random forest did no better than the logistic regression** in cross validation (accuracy 0.879 against 0.880), so the simpler model is kept.
8. **Employment status and home ownership are real but very small effects** (Cramer's V of 0.044 and 0.036). The model is just as accurate without them.
9. **The model does not use age, but it still approves younger applicants less often.** Part of the gap comes from lower income and credit scores, but not all of it, so group checks should be repeated on every new version of the model.
10. **The overall approval rate is 23.9%**, so most applications (76%) are rejected.
11. **The dataset is synthetic.** The findings describe how this data was generated and not how real lenders behave, so the model should be tested on real application data before anyone relies on it.

What to do next

1. Choose the cutoff from the real cost of a bad loan against a lost customer instead of the default 0.5, and send borderline cases (probability of approval between 0.3 and 0.5) for manual review.
2. Add the remaining credit columns (credit card utilization, bankruptcy history, previous defaults) and check whether the model improves.
3. Repeat the fairness checks for age, marital status and education on every new version of the model.
4. Test the approach on real application data.

Appendix A. Approval rate tables

Every group for every variable. Approval rate is approved loans divided by applicants.

Annual Income (USD)

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	15,000 to 31,676	5,000	52	1.0%
Lower middle	31,680 to 48,565	5,000	342	6.8%
Upper middle	48,567 to 74,388	5,000	1,163	23.3%
Highest 25%	74,400 to 485,341	5,000	3,223	64.5%

Loan to Income Ratio

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	0.03 to 0.26	5,000	3,431	68.6%
Lower middle	0.26 to 0.45	5,000	1,149	23.0%
Upper middle	0.45 to 0.78	5,000	184	3.7%
Highest 25%	0.78 to 7.54	5,000	16	0.3%

Total Debt to Income Ratio

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	0.02 to 0.18	5,000	3,374	67.5%
Lower middle	0.18 to 0.30	5,000	1,146	22.9%
Upper middle	0.30 to 0.51	5,000	236	4.7%
Highest 25%	0.51 to 4.65	5,000	24	0.5%

Loan Amount (USD)

Group	Range of values	Applicants	Approved	Rate
Smallest 25%	3,674 to 15,575	5,001	1,964	39.3%
Small middle	15,578 to 21,914	4,999	1,345	26.9%
Large middle	21,915 to 30,835	5,001	1,018	20.4%
Largest 25%	30,836 to 184,732	4,999	453	9.1%

Interest Rate

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	0.113 to 0.209	5,000	2,129	42.6%
Lower middle	0.209 to 0.235	5,000	1,368	27.4%
Upper middle	0.235 to 0.266	5,000	860	17.2%
Highest 25%	0.266 to 0.447	5,000	423	8.5%

Credit Score

Group	Range of values	Applicants	Approved	Rate
Below 450	343 to 449	396	55	13.9%
450 to 499	450 to 499	1,495	226	15.1%
500 to 549	500 to 549	4,129	801	19.4%
550 to 599	550 to 599	7,450	1,641	22.0%
600 to 649	600 to 649	5,809	1,736	29.9%
650 and above	650 to 712	721	321	44.5%

Assets to Liabilities Ratio

Group	Range of values	Applicants	Approved	Rate
Below 1 (owes more than owns)	0.01 to 1.00	4,629	849	18.3%
1 to 2	1.00 to 2.00	3,546	700	19.7%
2 to 5	2.00 to 5.00	5,129	1,204	23.5%
Above 5	5.00 to 559.49	6,696	2,027	30.3%

Employment Status

Group	Range of values	Applicants	Approved	Rate
Employed		17,036	4,089	24.0%
Self-Employed		1,573	438	27.8%
Unemployed		1,391	253	18.2%

Home Ownership Status

Group	Range of values	Applicants	Approved	Rate
Mortgage		7,939	1,997	25.2%
Other		2,036	417	20.5%
Own		3,938	980	24.9%
Rent		6,087	1,386	22.8%

Job Tenure (years)

Group	Range of values	Applicants	Approved	Rate
0 to 1 yrs	0 to 1	804	180	22.4%
2 to 3 yrs	2 to 3	4,434	1,041	23.5%
4 to 5 yrs	4 to 5	7,137	1,748	24.5%
6 to 10 yrs	6 to 10	7,374	1,748	23.7%
10+ yrs	11 to 16	251	63	25.1%

Savings Account Balance (USD)

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	73 to 1,541	5,000	1,140	22.8%
Lower middle	1,542 to 2,986	5,001	1,216	24.3%
Upper middle	2,987 to 5,873	4,999	1,214	24.3%
Highest 25%	5,874 to 200,089	5,000	1,210	24.2%

Appendix B. Core code

The main steps of the notebook. The full code, with every chart, is in `Loan_Approval_Financial_Risk_Analysis.ipynb`.

Load, clean and select columns

```
loans_data = pd.read_csv('Loan.csv')
loans_data['ApplicationDate'] = pd.to_datetime(loans_data['ApplicationDate'])

columns_ = ['ApplicationDate', 'AnnualIncome', 'CreditScore',
            'EmploymentStatus', 'LoanAmount', 'SavingsAccountBalance',
            'TotalAssets', 'TotalLiabilities', 'JobTenure',
            'HomeOwnershipStatus', 'LoanApproved']
loans = loans_data[columns_].copy()
```

Approval rate for each group

```
def approval_summary(data, category_col):
    summary = data.groupby(category_col, observed=True)['LoanApproved'].agg(
        Applicants='size', Approved='sum', ApprovalRate='mean')
    summary['ApprovalRate'] = (summary['ApprovalRate'] * 100).round(1)
    return summary

loans['Income Category'] = pd.qcut(loans['AnnualIncome'], q=4,
    labels=['Lowest 25%', 'Lower-middle', 'Upper-middle', 'Highest 25%'])
print(approval_summary(loans, 'Income Category'))
```

New columns

```
loans['LoanToIncome'] = loans['LoanAmount'] / loans['AnnualIncome']

liabilities = loans['TotalLiabilities'].replace(0, np.nan)
loans['AssetLiabilityRatio'] = loans['TotalAssets'] / liabilities
```

Logistic regression

```
X = pd.get_dummies(loans[numeric_features + categorical_features],
    columns=categorical_features, drop_first=True)
y = loans['LoanApproved']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

model = Pipeline([('scale', StandardScaler()),
    ('logreg', LogisticRegression(max_iter=1000))])
model.fit(X_train, y_train)
```

Cross validation and random forest

```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
forest = RandomForestClassifier(n_estimators=200, random_state=42, n_jobs=-1)

models_to_compare = {'Logistic regression': model, 'Random forest': forest}
for name, candidate in models_to_compare.items():
    scores = cross_validate(candidate, X, y, cv=cv,
                            scoring=['accuracy', 'roc_auc'])
    print(name, scores['test_accuracy'].mean(), scores['test_roc_auc'].mean())
```

Chi-square test and confidence interval

```
table = pd.crosstab(loans['Savings Category'], loans['LoanApproved'])
chi2, p_value, dof, expected = chi2_contingency(table)
cramers_v = np.sqrt(chi2 / (table.values.sum() * (min(table.shape) - 1)))

ci = binomtest(int(part.sum()), len(part)).proportion_ci(method='wilson')
```

Appendix C. Fairness check tables

Test applications only (4,000). Approved loans found = the share of loans that were really approved which the model also approves.

Marital status

Group	Applicants	Actually approved	Model approves	Approved loans found
Divorced	540	23.0%	21.7%	72.6%
Married	2,037	24.0%	21.3%	71.8%
Single	1,251	24.1%	18.7%	61.6%
Widowed	172	23.8%	20.3%	65.9%

Education level

Group	Applicants	Actually approved	Model approves	Approved loans found
Associate	820	19.8%	18.5%	69.1%
Bachelor	1,217	26.7%	22.2%	68.9%
Doctorate	159	43.4%	30.8%	66.7%
High School	1,154	14.4%	14.6%	71.7%
Master	650	36.0%	27.5%	65.4%

Employment status

Group	Applicants	Actually approved	Model approves	Approved loans found
Employed	3,402	24.2%	20.7%	68.8%
Self-Employed	311	26.4%	24.8%	72.0%
Unemployed	287	17.4%	13.2%	56.0%

Home ownership

Group	Applicants	Actually approved	Model approves	Approved loans found
Mortgage	1,624	24.9%	22.7%	70.1%
Other	415	17.3%	14.9%	66.7%
Own	786	26.5%	22.6%	72.6%
Rent	1,175	23.1%	18.0%	63.1%